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Hawkeye- Learn, Play, and Secure :An Artificial Intelligence Driven Gamified approach to Cyber Awareness

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Abstract. With the sudden rise of cyberattacks recently, cybersecurity awareness would be the need of the hour. Education in its recent forms cannot get users engrossed or relevantly interested. Hence, we proposed the concept of Hawkeye- a cybersecurity awareness website based on Human Computer Interaction principles. Four of the cyberattacks chosen would be most markedly beneficial to the users that would be communicated through the interface of Hawkeye in highly involving, situation-based games. The proposed solution employs a gamified approach to cybersecurity. Finding reveal that the use of interactive learning of proposed solution improves both interaction of the users with material being learned and knowledge retained. Integration with real world data and the live cyber news updates this would further enable cyber awareness and prevention

Keywords: Artificial Intelligence, Cybersecurity, Human-Computer Interaction, awareness, Hawkeye, gamification.

1 Introduction

Digital technologies have dramatically changed interaction, communication, and business operation but also increased the danger of cyber threats. Advanced attacks such as voice phishing or vishing, identity theft, malware, and deepfakes are now the most popular attacks to manipulate individuals and organizations. Voice phishing[1] uses some social engineering techniques to extract sensitive information over a call, and identity theft includes unauthorized access to accounts or fraud. Malware [2] is the most common dangerous attack, including variants like ransomware and spyware that have caused devastating harm. Deepfakes [3] are AI-generated media that manipulate sound or vision, spread misinformation and helps in complex frauds. But in fact, most traditional cybersecurity awareness platforms lack the interaction and engagement required for a truly effective learning experience; these are mostly static articles and

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quizzes. These strategies will not be able to make users interact actively; most importantly, they cannot fight with new threats like deepfakes. This delay makes them ill-prepared for cyber risks that are emerging nowadays. Therefore, a dynamic approach is needed.

Hawkeye is a cybersecurity awareness website, developed with principles of Human-Computer Interaction (HCI), in order to counter these challenges [4].

It has integrated interactive training materials like gamification scenarios and quizzes. Users will experience real practice for actual responses to imminent threats. In addition to the core functionalities, there is a community blogging facility for sharing user-experience- and strategy-sharing. This breeds community and experiential training.

Another live cyber news feed offers real-time updated information on emerging threats with the problem of outdated information many are experiencing in most of their platforms. Hawkeye's innovative design will focus on HCI principles in engagement, knowledge retention, and readiness for evolving threats; hence, it combines interactive learning and user-generated content with real-time updates to provide an even more holistic approach in educational cybersecurity that will allow students to better identify and then respond to cyber threats better.

2 Literature Review

2.1 Human-Computer Interaction (HCI) and User-Centered Design in Existing Systems

Most existing cybersecurity awareness platforms rely on traditional approaches such as information articles, instructional videos, and basic quizzes. The approach is based on some of the earliest principles developed in Human-Computer Interaction (HCI) that place emphasis on usability and simplicity [5]. It has been found that such principles make the access to the system much easier, but it usually fails to provide the necessary engagement in learning. As the Norman's Principles of Design, for a platform to be successful, it should provide intuitive interfaces, clear feedback mechanisms, and engaging user interactions. However, most of the existing systems do not advance from basic usability to active participation by the users [6].

For instance, although sites like StaySafeOnline and CIS Security Awareness demonstrate some essential principles of HCI, which include consistency and error prevention, they omit other complicated principles such as user control, gamification, and interactive feedback. Therefore, although users are receiving information, they are not actually involved in the process of application and practice [7].

On the other hand, User-Centered Design, as proposed by Gould and Lewis, requires that the involvement of users is present in every step of development so that design decisions can be aligned with specific needs and preferences. However, the existing systems are more content-driven rather than user interaction [8]. For example, Cybrary provides well-structured courses and lessons but misses chances to enhance interactive learning through real-world simulations and challenges.

2.2 Cybersecurity Awareness and the Limitations of Existing Approaches

Cybersecurity awareness forums are intended to empower the user with proper and actual knowledge and skills for better detection and response towards cyber threats. Normally, awareness forums would brief users regarding types of attacks like phishing, the theft of identity, malware, and the need to practice safe online behavior [9]. However, many of these depend heavily on passive learning tools such as informational articles, pre-recorded videos, and simple quizzes, which are not rich in engagement required for deep learning. Such approaches may deliver information well but fail to actively involve users in learning scenarios that reflect real-world threats [10].

While traditional awareness platforms cover well-known types of attacks, they often miss newer and more complex cyber threats, such as deepfakes and voice phishing. This focus on basic information delivery limits their relevance because users may not feel prepared to recognize or handle emerging cyber risks. Besides, most theory-based platforms lack the practicalities of application, hence find it challenging to ensure that behavioral change lasts. There tends to be a tough time in translating what they have learned into real life responses. Thus, cybersecurity awareness platforms should move away from the paradigm of simply content delivery and engage in interactive, hands-on learning for the preparation of present and future cyber threats [11].

2.3 Addressing the Research Gaps with Hawkeye

The limitations in existing systems such as static content delivery, lack of interactivity, narrow threat coverage, and inadequate community engagement—highlight the need for a more comprehensive approach to cybersecurity awareness. To address these issues, Hawkeye integrates theoretical foundations from HCI, experiential learning, and social learning to create a dynamic platform that emphasizes engagement, practical learning, and real-world relevance [12]. Key elements of this approach include:

- 1) **Interactive Games:** Drawing on Experiential Learning Theory, Hawkeye incorporates situation-based games that allow users to participate actively in realistic scenarios involving voice phishing, identity theft, malware, and deepfakes. This enables users to experiment with decision-making in a safe, controlled environment, promoting better retention and practical skills.
- 2) **Community-Driven Content:** Inspired by Social Learning Theory, Hawkeye offers a community blog where users can share personal stories and strategies. This feature not only strengthens community engagement but also provides a structured way to integrate peer experiences into the learning process.
- 3) **Real-Time Updates and Comprehensive Coverage:** Unlike existing systems that focus on limited threats or provide outdated information, Hawkeye includes a live cyber news feed to keep users informed of the latest developments in cyber threats. This real-time awareness helps users stay prepared for emerging issues.

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3 Methodology of the proposed system

The development process for the Hawkeye cybersecurity awareness website is user-centered. Data collection, through interviews and surveys, is done first to understand the needs and preferences of users. These are then condensed into a persona and an empathy map to show a clear picture of who the target audience is. All these findings lead to a user-friendly and interactive wireframe and gamified strategy. Finally, usability is assessed through a cognitive walkthrough so that the principles of HCI are strictly followed in the development to make the platform easy to use and effective for cybersecurity education. This entire process is represented in the block diagram in Fig. 1 which explains the flow from data gathering to final evaluation.

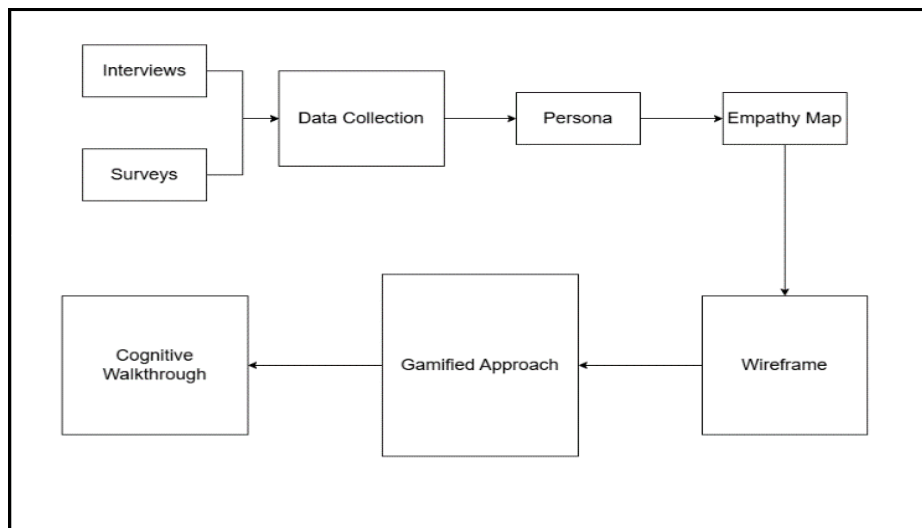


Fig. 1. Proposed Solution Block Diagram

3.1 Survey and Interview Result and Interpretation

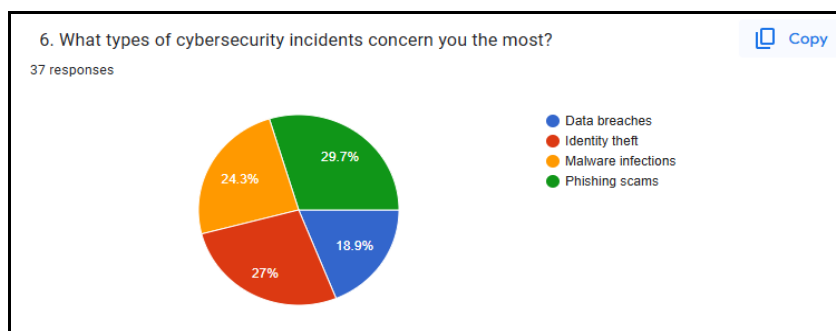


Fig. 2. Cybersecurity Threats concerns faced by users

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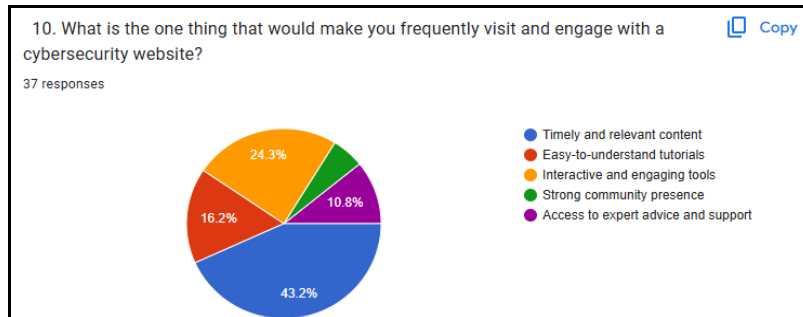


Fig. 3. What would make users visit and engage with a cybersecurity website?

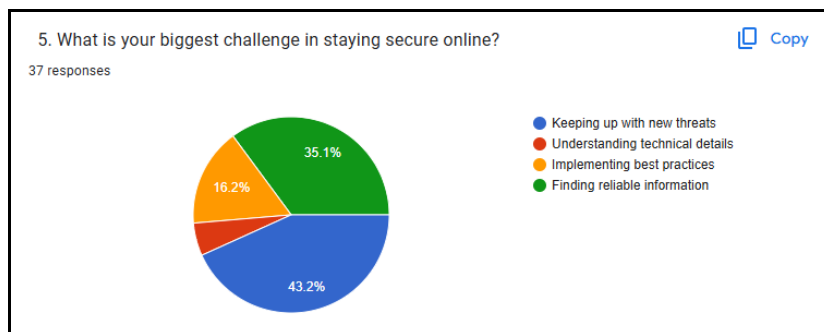


Fig. 4. Biggest challenge for users in staying secure online

Fig 2, 3, 4 describes the Survey and Interview Result and Interpretation

•**Older Users would prefer text content:**

For example, the older the user, say above 80, the more he or she would want the information in text form than interactive. They would want tutorials to be easy to read and comprehend with clear instructions, just as observed in the second image by the desire to have easy-to-understand tutorials standing at 43.2 percent [21].

•**Younger Users Below 30 Like Gamification of Learning:** Users below 30 years like to have interactive and gamified tools in order to learn about cybersecurity. That is indicated by 10.8% who are interested in using interactive tools for more frequent engagement with cybersecurity websites. These gamified methods may make them understand the threats-also in phishing scams and malware-much better [23].

•**High interest in the emerging threats across demographics:** It would be very apt to state that, overall, every demographic shows high interest in phishing scams at 29.7%, identity theft at 27%, and malware infections at 24.3%, as

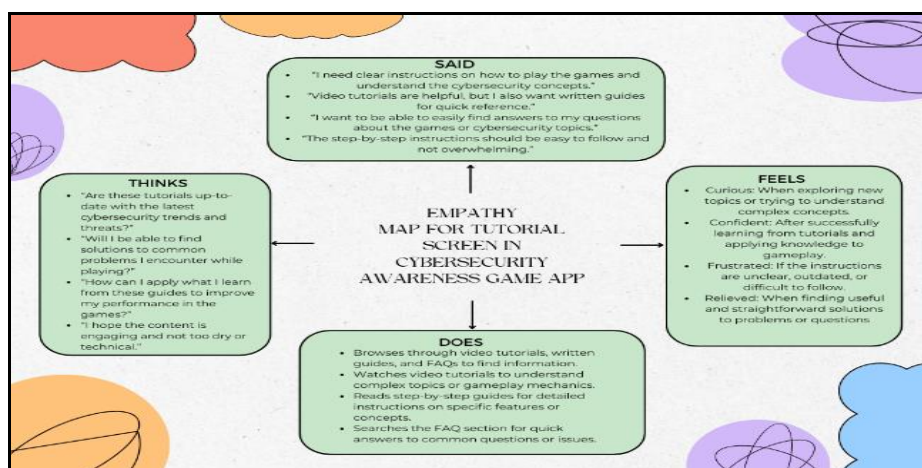
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captured in the first image. In other words, they show interest in being timely notified and renewed with refreshed content regarding new threats [22].

- **The victims of cyberattacks are in the 40-60 age bracket:** Indeed, comprehensive and practical resources would be required by the age group of 40-60 as this age bracket has encountered more cyberattacks. In fact, they cherish getting good information; 35.1% chose "Finding reliable information" as one of the most difficult parts of staying secure online [23].
- **Participants aged between 30 and 50 years prefer video tutorials:** The video tutorial and webinars summarize complicated topics in an easily understandable form, which is in tandem with the choice of easy-to-understand tutorials 43.2% as a factor for frequent engagement with cybersecurity website as shown in the image below [21].
- **Community Involvement:** Users under 40 would like to shared experiences and advice, a fact that can be asserted by the evidence that 16.2% believe strong community presence is key to staying engaged on cybersecurity platforms.

3.2 Empathy Map

This empathy map outlines the important needs, thoughts, feelings, and actions of the users interacting with the tutorial screen within the cybersecurity awareness game application. The mapping discussed in Fig 5 captures a need for well-defined, clear, and easy-to-access instructions as well as an understanding of how to protect themselves in an ever-evolving world of cybersecurity knowledge, which translates into the video guides and text guides. Users should be able to trust the game for applying cybersecurity concepts and should find simple, step-by-step instructions that are less frustrating and more engaging. This empathy map forms the basis for developing an intuitive tutorial experience that is both informative and supportive of users' learning goals [13].



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Fig. 5. Empathy Map

3.3 Persona

These personas provide insight into the diverse needs and challenges of Hawkeye users. By capturing different age groups, tech familiarity levels, and cybersecurity concerns, these profiles help guide the platform's design to be both interactive and accessible, ensuring that users like Anni and Kurkure in Fig 6 and Fig 7 find content that meets their learning styles and security priorities [14].

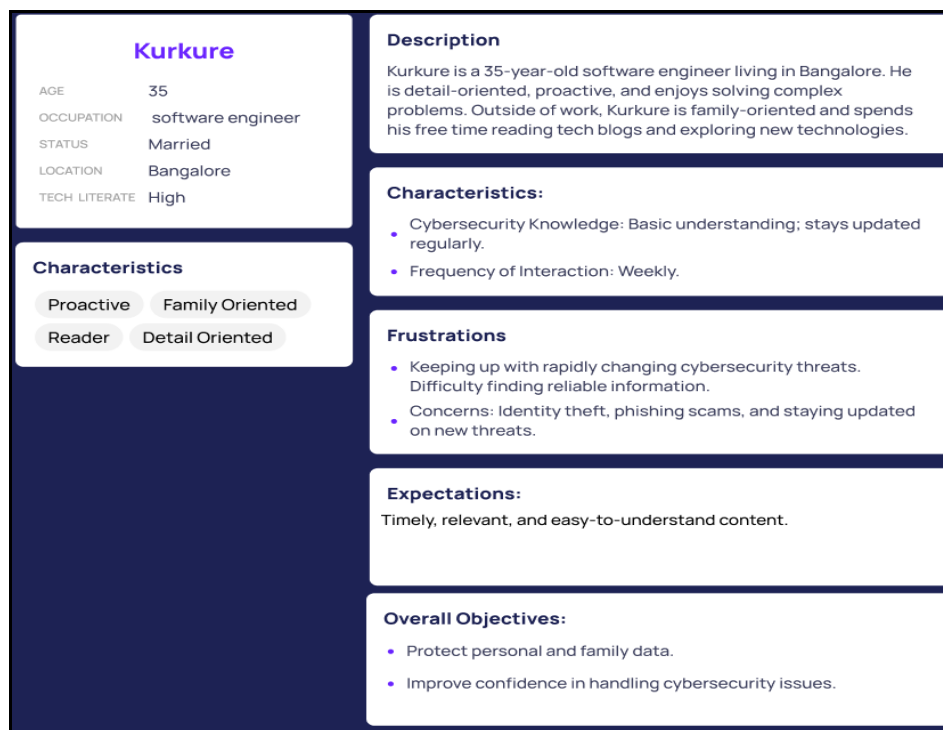


Fig. 6. User Persona

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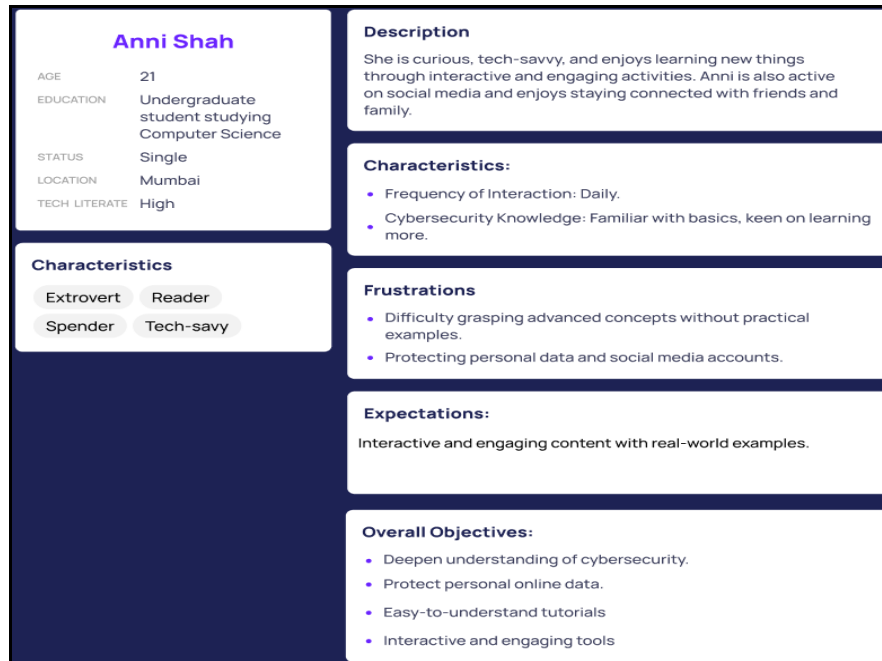


Fig. 7. Persona of Anni Singh

3.4 AI in HCI for Cybersecurity Awareness Website

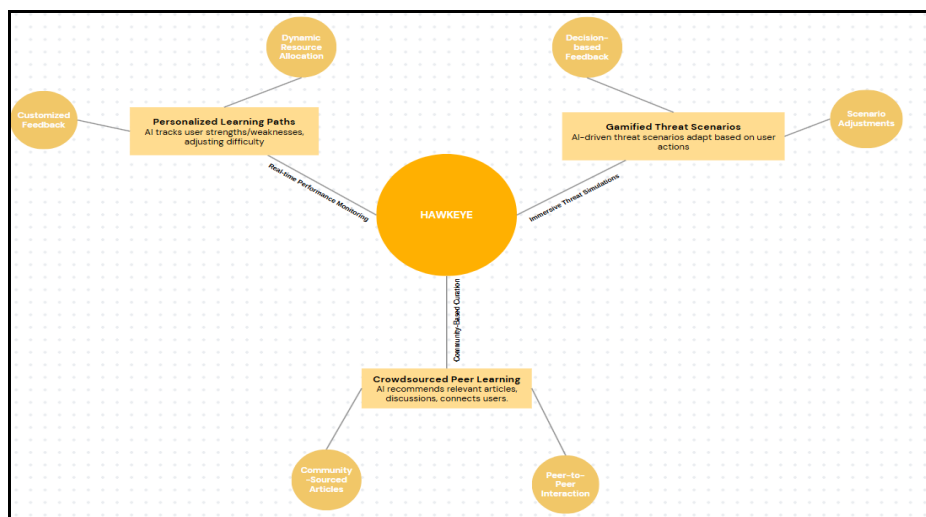


Fig. 8. Flowchart of AI Integration in Hawkeye

1) Personalized Learning Paths

This is how the feature of Personalized Learning Paths brings AI in performance monitoring and then adjusts delivery according to the workings [26][27]. Through quizzes and exercises, AI continually analyzes and learns through the areas of strength and weaknesses as it dynamically alters the degree of difficulty so that material remains suitably challenging. In each activity, the software outputs specific self-assessment feedback that is matched with target recommendations and resources, which can help highlight some of these weaknesses [25]. As flowing learning is inculcated interactively to his needs as the users are working through the program in real-time, there are continuous spikes in performance, engagement, retention, and the mastery of content [28][29].

2) Gamified Threat Scenarios

In the gamified simulations of Hawkeye, full immersion participation is realized, especially in highly realistic attack scenarios such as phishing and so forth. They are real replicas of threats, a live simulation [30]. Real-time feedback gives the user marks on decision making, pointing out mistakes as well as strategies that would work well in their improvement responses. This gamification will allow practice to happen within safe parameters, holding their interests alive by keeping them engaged through interactive scenario-based learning [28].

3) Crowdsourced Peer Learning

Crowdsourced peer learning employs AI for offering relevant articles, blog posts, and updates to users based on the activities they perform. It further scans and curates' community contents ensuring only quality relevant information for connected users with similar interest or a problem [24]. Encouraging people to collaborate and learn together is achieved by trying to connect people interested or having a similar type of problem which creates dynamic communities where users share their knowledge and exchange discussions and hold abreast with the trends that are current [29][30].

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4 Discussion and Result

4.1 Interface

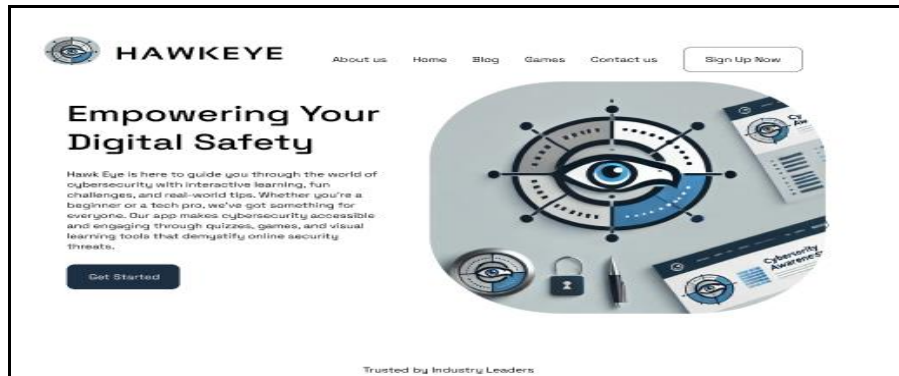


Fig. 9. Start Page

The Home Page in Fig 9 becomes a portal into Hawkeye. It fulfills the website's mission-to arm its users with knowledge about digital safety through hands-on learning. A "Get Started" button gets newcomers started, and a clean design full of easy navigation links steers visitors to key sections like Blog, Games, and User Stories.

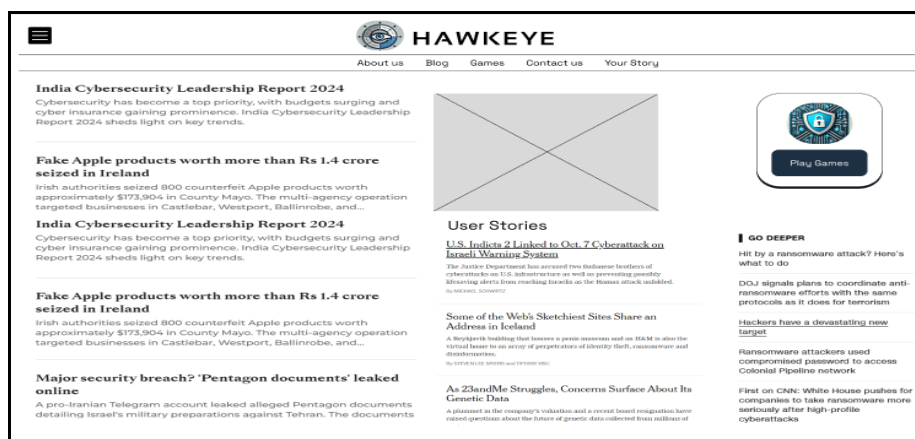


Fig. 10. Home Page

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Fig 10 describes the News Page, users will always be informed of the latest information regarding cybersecurity news and trends. The page includes always features like "Top Targeted Countries" and "Prevention Starts Here" in the desire to provide updates and in-depth guidelines.

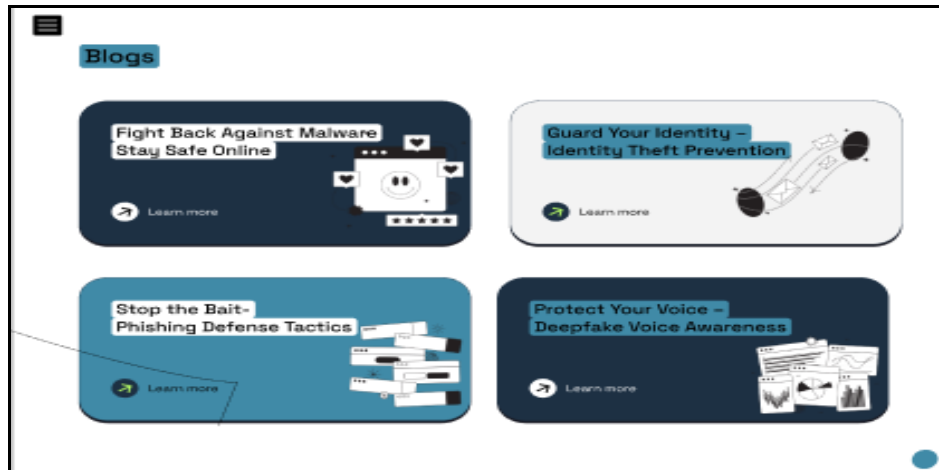


Fig. 11. Blog Page

The Blog Page in Fig 11 and 12 has various issues concerning cybersecurity. It has advisory posts such as "Malware", "voice phishing", "Deepfake" and "Identity theft", informing users on the major cyber threats and information protection strategies.

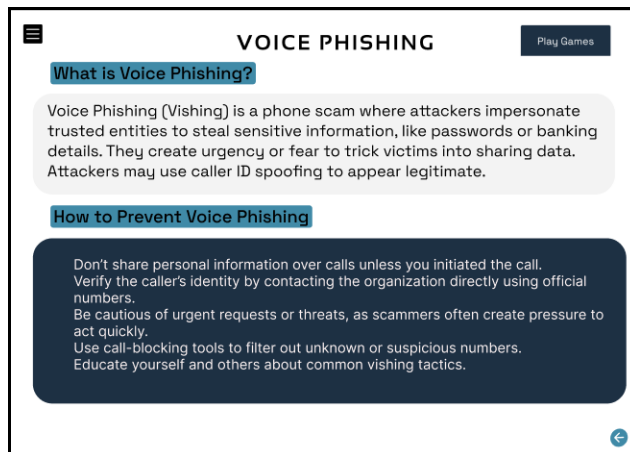


Fig. 12. Blog Page

These pages have a clean and consistent design with clear headings, simple explanations, and actionable prevention tips. Each page starts with a title and a

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brief, clear definition of the cyber threat, introducing the concept via a "What is" section, followed by a "How to Prevent" section in concise bullet points for easy reading. It applies strong headings, boxes containing tips with contrasting color for backgrounds, and minimalist icons or buttons like "Play Games" to attract users in taking part.

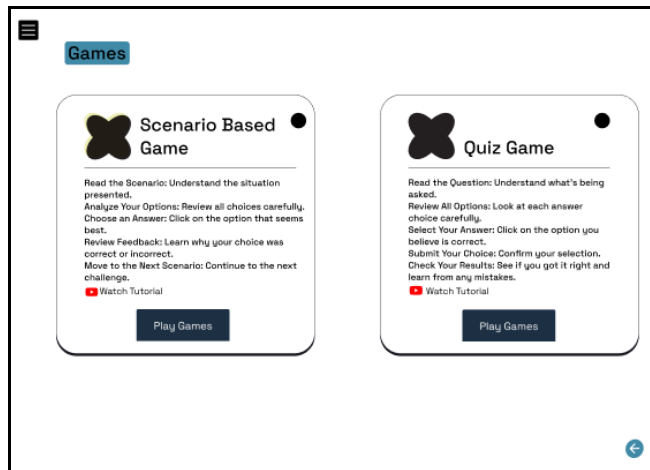
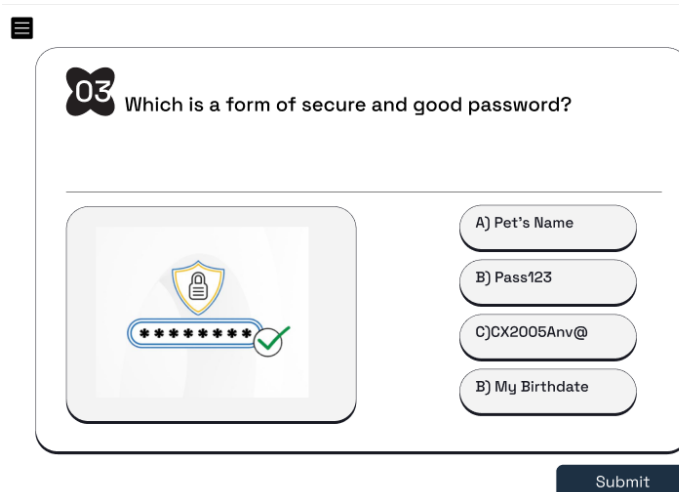


Fig. 13. Game Page

Fig 13 describes the Games Page provides students with the choice of two interactive modes: Scenario-Based Games and Quiz Games. For each of these game types, it provides brief instructions, encouraging students to learn better by playing.



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Fig. 14. Game Page

- These images represent the different interactive scenario-based games and quizzes aimed at teaching people about cybersecurity threats and how to be safe. For each scenario, there will be a real-life situation, such as receiving an unknown message or an unrecognizable email. Different options are presented, with the user selecting the most appropriate response. The game then immediately responds with which is the right answer with explanations. This interactive design would allow users to build hands-on skills by engaging with real-life decision-making and enforcing knowledge through clear, simple explanations. The clean intuitive layout encourages the user



The image shows a web interface for a 'User Blog'. At the top, there is a hamburger menu icon and the title 'User Blog'. Below the title is a blue button labeled 'Upload news'. The main content area is a light gray box containing a form with three input fields: 'Name', 'Email*', and 'Message*'. Each field has a placeholder text: 'Name', 'Email', and 'Message'. Below the 'Message*' field is a dark blue button labeled 'Send Message'. To the right of the form is a decorative graphic featuring a large black star and a smaller blue star, both with radiating lines. A small blue circular arrow icon is located at the bottom right of the form area.

Fig. 11. User stories Page

- The User Stories Page enables users to share their own experiences. A simple input form enables users to enter their name, email, and message in order to contribute stories or updates regarding their cybersecurity experiences.

4.2 Cognitive Walkthrough

A cognitive walkthrough is a usability evaluation that focuses on understanding user interaction with any system when trying to do a given task. In a cybersecurity-related context, a cognitive walkthrough could help in studying whether features can be understood intuitively by users to identify relevant security options that may call

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for an adequate response upon receiving security alerts or prompts. This approach identifies the potential usability issues that might prevent users from taking safe actions, understanding alerts, or learning about cyber threats effectively by examining each step of a user's journey through a cybersecurity platform. Applications like cybersecurity awareness Hawkeye would necessitate cognitive walkthroughs so that users could find their relevant information but engage in more interactive elements, like playing games, reading news articles, or blog posts, to enhance security knowledge and a decision-making skill [18].

- Games Page

Is the effect of action the same as users' goal?

Yes, on the Games Page, when users click the "Play Games" button they immediately go to the interactive game experience, which is consistent with their goal of either engaging in scenario-based learning or quiz games about cybersecurity threats. This configuration meets the expectation for easy access to the games, supporting the learning process.

Will users see that action is available?

Yes, the game options and "Play Games" buttons are right up front on the page for visual clues as to how to start a game. Each game type is self-explanatory, so users will instantly know which game they'd like to play.

Do users know this is the action they want?

Yes, the labels such as "Scenario-Based Game" are quite obvious and clearly describe the kind of gaming experience being provided. The descriptive label enables users to pick the correct game based on their learning style and therefore assures them that they are making the right choice.

Will users know what the feedback is about?

Yes, the games give immediate feedback after each question that shows explanations of correct or incorrect answers. This immediate feedback not only confirms that the action they took is a success-that is, their answer is correct-but also reinforces learning by providing explanations for each correct answer. In this way, it encourages users to improve and keep playing.

- News Page

Is the effect of action the same as users' goal?

Yes, users will visit the News Page to read news articles on cybersecurity and when they click on any headline of the news, a full article view will open. This will be aligned with the goal of the user to keep up-to-date with the latest cybersecurity trends and threats.

Will users know that action is available?

Well. Articles are listed with clear headlines and "Read More" links. The format lets the user see right away that each headline is actionable and brings them to read an article in full.

Will users know the action is the one they need?

Yes, clicking on a news headline or on a "Read More" link obviously will open the full article, which satisfies their need to access more detailed news content. Interface cues such as underlined text or a particular link styling further support this understanding.

Will users understand the feedback they get?

That's right, but by clicking on a headline or link, users will go directly to the full article. Such a direct transition confirms to the user that their action was successful as it reminds them that he or she accessed the content.

- Blog Page

Blog Page Is the effect of action the same as users' goal?

Yes, on the Blog Page, users who want to learn more about something by clicking "Learn More" on a blog post card can see the detailed information about that post, which aligns with them wanting to read in-depth, detailed blog content on cybersecurity issues.

Will users see this action is available?

Yes, blog cards are presented in an orderly fashion, with a clear "Learn More" link on each one. Visually, it is obvious that when the "Learn More" is clicked, it will lead further to content.

Will the users know that the action is the one they need?

Yes, the "Learn More" label intuitively cues that more information is available. The label is familiar to users as a signal to access additional detail, which will take them to the full blog post.

Will users understand the feedback they receive?

Yes, clicking the "Learn More" will send users directly to the full blog article and give them immediate feedback that they successfully accessed the post they wanted. Thus, through direct navigation, confirmation of the action is presented, along with assurance that the click executed what was desired.

- User stories Page

Is the effect of action the same as users' goal?

On the User Stories Page, users clicking on a story title will read experiences of others, and those who click "Upload News" will be brought to a form to tell their own story. That's a reflection of the two goals the users have, either reading peer experiences or contributing their own.

Do users understand that action is available?

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Yes, the story titles are prominent and the "Upload News" button is front-and-centre, so these acts are easily discoverable. This makes it obvious to users how either to browse existing stories or share their own.

Will users know the action is the one they need?

Yes, the tag "Upload News" literally tells one how to include new content; and titles of stories which are hyperlinks point out full stories. Such intuitive names mean that users will know to click on an action appropriate to their intention to read or contribute.

Will they understand the feedback they receive?

Yes, once users have uploaded a story through "Upload News, they get confirmed that their submission has been successfully uploaded, with the message indicating that. Such a confirmation reassures the users that the system had received their input and closure to the action is provided, making it much more enjoyable to the users.

5 Conclusion

Through the design, the evolution of Hawkeye really helped develop Human-Computer Interaction principles [19] that would truly strengthen awareness of security awareness, especially with very interesting user-centered designs. Usually, traditional methods fail in providing any form of prolonged engagement among users as they hardly try to bring aspects to reality. In dealing with the limitations above, this platform focuses on such elements that interact with learners more. It covers all the critical areas of threats, including voice phishing, identity theft, malware, and deepfakes. This means equipping all users with age-specific, practical knowledge and hands-on experience. The website includes user-generated stories and live news updates regarding evolving threats. The success of Hawkeye shows that HCI-based platforms are the key to closing the gap between theory and practice, providing an all-encompassing, entertaining, and interacting learning experience. Still, by providing constant new versions and feedback from its users, Hawkeye will always change, keeping its users up to date with fast-changing new digital paradigms.

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5.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraphs that follows a table, figure, equation etc. does not have an indent, either. Subsequent paragraphs, however, are indented.

Sample Heading (Third Level). Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Forth Level). The contribution should contain no more than four levels of headings. The following Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1 st -level heading	1 Introduction	12 point, bold
2 nd -level heading	2.1 Printing Area	10 point, bold
3 rd -level heading	Run-in Heading in Bold. Text follows	10 point, bold
4 th -level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \quad (1)$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

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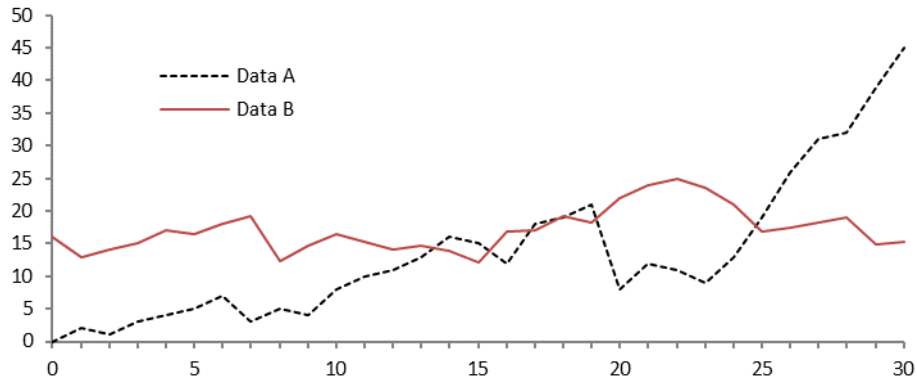


Fig. 1. A figure caption is always placed below the illustration. Short captions are centered, while long ones are justified. The macro button chooses the correct format automatically.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], as well as a URL [5].

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